

Two-Term Boundary Complex

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Summary

Pass 85 compares three two-term models for the finite-prime phantom boundary:

$$C_{\mathbb{Z}} = [\mathbb{Z} \rightarrow \widehat{\mathbb{Z}}], \quad C_{\mathbb{R}} = [\mathbb{R} \rightarrow \Sigma], \quad C_{\mathbb{Q}} = [\mathbb{Q} \rightarrow \mathbb{A}_f],$$

where

$$\Sigma = (\mathbb{R} \times \widehat{\mathbb{Z}})/\mathbb{Z}, \quad \epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}.$$

All three complexes have quotient ϵ , but they do not preserve the same extension data.

Main Claims

- **Same quotient.** Each complex has injective differential and $H^1 \cong \epsilon$.
- **Finite/Hausdorff acyclicity.** At every finite modulus the diagonal image is all of $\mathbb{Z}/N$, so ordinary finite cokernels vanish. Likewise, $\mathbb{R}$ is dense in Σ and $\mathbb{Q}$ is dense in $\mathbb{A}_f$.
- **Shear-preserving pushout.** The comparison $C_{\mathbb{Z}} \rightarrow C_{\mathbb{Q}}$ is the pushout along $\mathbb{Z} \hookrightarrow \mathbb{Q}$ and preserves the finite-adele extension

$$0 \rightarrow \mathbb{Q} \rightarrow \mathbb{A}_f \rightarrow \epsilon \rightarrow 0.$$

The archimedean row $C_{\mathbb{R}}$ has the same quotient but not this Borel shear class.

Machine Verification

code/scripts/check-pass85.py produced artifacts/reports/pass85-two-term-boundary-complex-check.json with overall PASS. It checks dense finite shadows, the non-Mittag-Leffler lcm kernel tower, compatible unit residues, and the comparison table distinguishing the shear-preserving finite-adele pushout from the archimedean row.

Next Use

Pass 86 should state the universal property of $C_{\mathbb{Q}} = [\mathbb{Q} \rightarrow \mathbb{A}_f]$: among quotient models of ϵ with divisible kernel, it should be the initial receiver of $C_{\mathbb{Z}}$ that preserves the unit/shear class and kills Hausdorff finite cokernels.